

1 Measurements of mass and volume

EXPERIMENT 1

The scientific method

Goal

The goal of this experiment is to understand how the scientific method works. To do this you will make observations, look for patterns and come up with a hypothesis to *identify an unknown compound*. You will do this by comparing the reaction of the unknown compound with a set of known-reference-compounds, also called blanks. In this experiment, you will obtain some blank chemicals and an unknown. Your unknown is the same as one of the blanks. You will test each of these chemicals with a set of reagents. You have to observe the result of the test and record any change, such as *gas evolution*, the formation of a *precipitate*, a *change of color*, or maybe *no reaction*. Simultaneously, you will run similar tests for the unknown sample that is identical to one of the blanks. You will determine the unknown's identity by matching its characteristic reactions with one of the blanks.

Materials

- ☐ Six test tubes.
- ☐ Unknown compound.
- ☐ The blanks: NaCl(s), NaI(s), Na₂CO₃(s), Na₂HPO₄(s) and Na₂SO₄(s).
- ☐ The reagents: HNO₃(aq), Ba(NO₃)₂(aq), AgNO₃(aq) and thymol blue.

Background

Procedure

Part A: Indicate your hypothesis

- ☐ *Step 1:* – Obtain the unknown compound, the blanks, and the reagents.
- ☐ *Step 2:* – Compare the unknown with the blanks and guess which blank looks like the unknown. That is your hypothesis. Write down the name of your guess on the results page.
- ☐ *Step 3:* – Wash a 100 mL graduated cylinder and 400 mL beaker. Place a pea-size portion of the unknown in the beaker, and add 150 mL of water using the cylinder to measure it. Swirl until all the solid has been totally or partially dissolved.
- ☐ *Step 4:* – Set aside the remaining portion of your unknown.

Good Lab Practice

 Obtain directions from your lab instructor for discharging the solutions that you will use in this experiment.



Part B: Testing for gas

- ☐ *Step 1:* – In this first experiment you will use the solid version of the blanks, not the droppers. Obtain six test tubes. In each test tube, you will add a blank reagent in a solid state, that is, one blank per test tube. To each test tube, you will add a pea-size sample of a different blank in a solid state using its spatula. You will add the unknown in the sixth test tube. This way, place a pea-size solid sample of NaCl in the first test tube. Add a pea-size sample of NaI in the second test tube, and so on with the rest of the blanks, Na₂CO₃, Na₂HPO₄, Na₂SO₄ and your solid unknown in the following test tubes. Mark your test tubes for recognition.
- ☐ *Step 2:* – Add five drops of HNO₃ ⚠ to each test tube and record your observations in the results table: *GE*, for gas evolution, *NGE* for no gas evolution.
- ☐ *Step 3:* – Discharge the solutions from the test tubes into the right waste container. Wash the test tubes, and rinse them with distilled water.

Part C: Testing for a barium precipitate

- ☐ *Step 1:* – For the following test will use only liquid samples in the dropper, not the powders. Use the solutions of NaCl, NaI, Na₂CO₃, Na₂HPO₄ and Na₂SO₄ found in the laboratory, and the solution of your unknown prepared in Step 3 for the remaining tests in this practice.
- ☐ *Step 2:* – Using six clean test tubes, add 20 drops of each of the solutions as well as the unknown to separate test tube. In short, five of the test tubes should have a different blank and the sixth test tube should have the unknown solution. Add 3 drops of a solution of ammonia (NH₃) in each test tube ⚠ and swirl until solved.
- ☐ *Step 3:* – Add five drops of the Ba(NO₃)₂ ⚠ solution to each of the test tubes. Shake gently to obtain homogeneity. Examine each test tube carefully and look at the results. Record your observations: *P* for precipitate, *NP* for no precipitate.

⚠ CAUTION!

- ⚠ Nitric acid (HNO₃) and ammonia (NH₃) ⚠ can cause chemical burns.
- ⚠ Solutions containing barium are toxic.

Part D: Testing for silver precipitate

- ☐ *Step 1:* – Using six clean test tubes, add 20 drops of each of the solutions including the unknown to separate test tubes (e.g. NaCl, NaI, Na₂CO₃, Na₂HPO₄, Na₂SO₄, and unknown). In short, five of the test tubes should have a different blank and the sixth test tube should have the unknown solution.
- ☐ *Step 2:* – Add five drops of the nitric acid AgNO₃ solution to each test tube. Record your observations: *P* for precipitate, *NP* for no precipitate. Include the color of the precipitate.

Part E: Testing with thymol blue

- ☐ *Step 1:* – Using six clean test tubes, add 20 drops of each of the solutions including the unknown to separate test tubes (e.g. NaCl, NaI, Na₂CO₃, Na₂HPO₄, Na₂SO₄, and unknown). In short, five of the test tubes should have a different blank and the sixth test tube should have the unknown solution.
- ☐ *Step 2:* – Add five drops of the thymol blue solution to each test tube. Shake each test tube gently before recording your observations: *write down Blue, Yellow green, or Red.*

Part F: Identifying the unknown compound

- ☐ *Step 1:* – The unknown is one of these compounds: NaCl, NaI, Na₂CO₃, Na₂HPO₄, Na₂SO₄. You should be able to identify your unknown compound by matching its reactions with those of one of the known samples.
- ☐ *Step 2:* – When done with the experiment, clean up all test tubes using a test tube brush and soapy water.

STUDENT INFO

Name: _____ Date: _____

Pre-lab Questions**The scientific method**

1. Classify the following compounds as *ionic, covalent, acids, bases* or *oxo salts*:

NH ₃	_____	SO ₃	_____
HNO ₃	_____	NaF	_____

2. Give the names or formulas for the following compounds:

NaCl	_____	HNO ₃	_____
Sodium iodide	_____	Sodium carbonate	_____

3. Name or give the formula of the following compounds:

SO ₂	_____	NO ₂	_____
Lithium fluoride	_____	Sodium oxide	_____

4. What does a pea size of solid mean?

5. Where in the lab you will be disposing of the disposals from this experiment?

6. The following table resulted from an experiment, very similar to the one you are going to carry. The goal was to identify an unknown compound between KCl and KI. Look at the reactivity patterns and identify the unknown.

		Reagents			
		HNO ₃	Ba(NO ₃) ₂	AgNO ₃	Thymol Blue
Blanks	KCl	NP	NP	P _(white)	Green
	KI	NP	NP	NP	Green
	Unknown	NP	NP	NP	Green

My unknown is = _____

STUDENT INFO

Name: _____ Date: _____

**Results
EXPERIMENT 1****The scientific method**

Unknown Label # = _____ Your guess = _____

		Reagents			
		HNO ₃	Ba(NO ₃) ₂	AgNO ₃	Thymol Blue
Blanks	NaCl				
	NaI				
	Na ₂ CO ₃				
	Na ₂ HPO ₄				
	Na ₂ SO ₄				
	Unknown				

(write GE for gas evolution, NGE for no gas evolution, P for precipitate, NP for no precipitate and Blue, Yellow or Green for the color for the Thymol blue experiment)

My unknown is = _____

STUDENT INFO

Name: _____ Date: _____

Post-lab Questions

The scientific method

1. Classify the following compounds as *ionic*, *covalent*, *acids*, *bases* or *oxo salts*:

Ba(OH) ₂	_____	CO ₂	_____
HCl	_____	H ₂ SO ₄	_____
LiCl	_____	AuCl	_____
Na ₂ SO ₄	_____	NaOH	_____

2. Give the names or formulas for the following compounds:

HNO ₃	_____	Sodium sulfate	_____
Barium nitrate	_____	Na ₂ CO ₃	_____
AgNO ₃	_____	Nitric acid	_____

3. Name or give the formula of the following compounds:

CuCl ₂	_____	Hydrogen chloride	_____
Nickel (II) chloride	_____	BaO	_____
CoO	_____	MgO	_____
Sodium iodide	_____	Potassium iodide	_____
NH ₃	_____	LiH	_____

